Linear Functions

A function is *linear* if its slope at every point is the same.

Equivalently, any linear function can be written in the form

$y = f(x) =$ _____,

where m is the _____ and c is the _____.

Preliminaries

Linear Functions from Points and Slope

- Suppose $g(x)$ is a linear function. Find its equation given that its slope is 6 and that $g(5) = 9$.

- Suppose $f(x)$ is a linear function. Find its equation given that $f(2) = 4$ and $f(4) = -2$.

Parallel and Perpendicular Lines

- Given two linear functions $f(x) = m_1x + c_1$ and $g(x) = m_2x + c_2$, how can you tell if the two lines are *parallel*?

- Suppose that the line l is parallel to $y = \frac{1}{2}x - 2$ and passes though the point $(1, 2)$. Find its equation.

- Given two linear functions $f(x) = m_1x + c_1$ and $g(x) = m_2x + c_2$, how can you tell if the two lines are *perpendicular*?

- Find an equation for the line l that is perpendicular to the line $y = \frac{1}{3}x + 2$ and passes through the point $(1, 2)$.

Fill in the blank: The slope, m, of a linear function is $\frac{\Delta y}{\Delta x}$. If $\Delta x = 1$, then $\Delta y =$ ___.

An Important Characterization of the Slope of a Linear Function:

Slope is the amount the _____ variable increases by every time the _____ variable increases by ___.

Modeling with Linear Functions

1. □Regrade One group calculated in the Cancer Mortality lab that if I is the index of exposure, and M is cancer mortality (in cancer deaths per 20,000 people per year), then a good model for the data is

$$M = 11.2I + 112.$$

If a mayor of a town manages to decrease the index of exposure in his town, whose population is 115,000 people, by exactly 1.7 units, how many cancer deaths does the model predict will be prevented over the next five years?

2. □Regrade **The Apple Orchard** In a certain orchard, the following facts are known:

- if 24 trees are planted per acre, each tree will yield 600 apples and
- for each additional tree planted per acre, the yield decreases by 12 apples per tree.

Let the number of trees per acre be x, and denote the number of apples per tree by A.

(a) What is A if x is 24 trees per acre? Include units.

(b) What information in the question tells you that A is a *linear function* of x? What is its slope?

(c) Find a formula for A in terms of x. Include units. (Hint: you already have a point and the slope!)

Proportionality

- We say that y is *directly proportional* to x if there is a non-zero constant k such that
$$y = kx;$$
- We say that y is *inversely proportional* to x if there is a non-zero constant k such that
$$y = \frac{k}{x}.$$

In either case, k is called the *constant of proportionality*.

3. Given a table of data for values x and y, briefly describe how you would check if:

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(a) y is directly proportional to x.

(b) y is inversely proportional to x.

4. The following law of gravitation was suggested by Ismael Bullialdus in 1645: "The gravitational attraction force (F) between two point masses is directly proportional to the product of their masses and inversely proportional to the square of their separation distance (d)." Assuming the masses are m_1 and m_2, write this as a formula.

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More Questions

5. A water pipeline follows the straight-line path $y = 2x - 1$. A farmhouse is located □Regrade at the coordinates $(4, 2)$. Find the exact shortest distance from the farmhouse to the pipeline.

   (Hints: The shortest distance occurs along a line perpendicular to the pipeline. The distance between two points (x_1, y_1) and (x_2, y_2) is given by $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.)

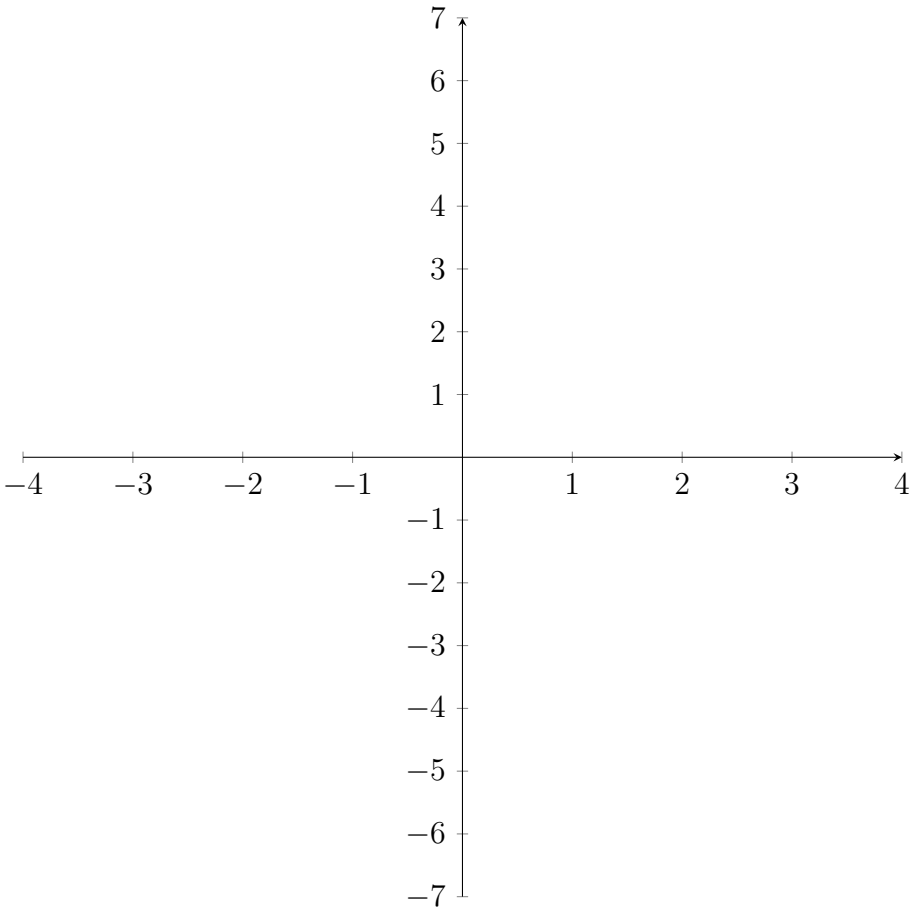
6. Given that a triangle ABC in the coordinate plane has its vertices at the points $(1, 4)$, □Regrade $(4, 8)$, and $(5, 1)$, show that triangle ABC is a right triangle. (Hint: If we have a right angle, then two of the sides are _____ to each other.)

7. Consider the following *piecewise linear* function:

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$$h(t) = \begin{cases} -2t - 1, & t \leq -2, \\ t + 5, & -2 < t < 1, \\ -0.5t + 0.5, & t \geq 1. \end{cases}$$

(a) Draw its graph.



(b) Fill in the blank: At $t = 1$, the function is not _____.